Report: Password Vault Local (PVL)

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Abstract

Password Vault Local (PVL) is a secure, offline, Python-based desktop application designed to manage, store, and retrieve user credentials safely. Utilizing robust cryptographic standards such as Fernet symmetric encryption and PBKDF2HMAC key derivation, PVL provides a localized, user-friendly environment for password management without relying on cloud services.

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1 Introduction

With the increasing number of digital accounts, managing passwords securely has become a necessity. Password Vault Local (PVL) addresses this need by providing a secure desktop application that stores passwords locally, ensuring that sensitive data never leaves the user's device. It features a modern graphical user interface (GUI) developed using `customtkinter`.

2 Core Functionalities

PVL supports a comprehensive set of functionalities for daily password management:

- **Master PIN Authentication:** Access to the vault is protected by a Master PIN. The application verifies the user upon launch before decrypting the data.
- **CRUD Operations:** Users can seamlessly Add (Create), View (Read), Edit (Update), and Remove (Delete) credentials.
- **Dynamic Search and Filtering:** Real-time filtering allows users to search entries by 'Source' or 'Category'.
- **Clipboard Integration:** One-click copying of usernames or passwords to the system clipboard for immediate use.
- **Visibility Toggle:** Users can securely mask or reveal the password in the input fields using an eye-icon toggle.

3 Key Features

The software distinguishes itself through several technical and UI/UX features:

- **Robust Encryption:** Uses Fernet (symmetric encryption) from the Python `cryptography` library to encrypt the vault data file.
- **Secure Key Derivation:** Derives the encryption key using PBKDF2HMAC with SHA-256 and a 16-byte random salt, making it highly resistant to dictionary or brute-force attacks.
- **Modern GUI:** Replaces standard Tkinter interfaces with `customtkinter`, providing a sleek, professional, and responsive design featuring rounded corners, custom fonts, and a spreadsheet-like Treeview layout.
- **Automatic UI Population:** Clicking on any entry automatically fills the input fields and switches the application into an 'Update' mode, enhancing user experience and workflow.
- **Data Masking:** Passwords are automatically masked in the data table to prevent shoulder-surfing.

4 System Requirements

To run PVL, the following software and libraries are required:

- **Operating System:** Windows, macOS, or Linux.
- **Python:** Version 3.7 or higher.
- **Dependencies:**
  - `customtkinter` (for the modern UI components)
  - `cryptography` (for Fernet, hashes, and PBKDF2HMAC)
- **Standard Libraries:** `os`, `json`, `hashlib`, `base64`, `tkinter`.

5 Architecture and Data Storage

PVL operates completely offline. Its file structure consists of:

- `vault_data/config.json`: Stores the hashed Master PIN (SHA-256) and the base64 encoded salt.
- `vault_data/vault.enc`: Contains the Fernet-encrypted JSON array of the user's saved credentials.

When the user sets up the vault, a salt is generated, the PIN is hashed, and both are saved. Subsequent logins verify the hashed PIN, combine the provided PIN with the salt to derive the Fernet key, and decrypt the vault file in memory.

6 Limitations

While highly secure, PVL has some inherent limitations:

- **No Cloud Backup/Sync:** Because it is purely local, if the device's hard drive fails or the `vault_data` directory is deleted, all passwords are irrecoverably lost unless the user manually backed them up.
- **No Password Recovery:** If the user forgets their Master PIN, the encryption key cannot be derived. There is zero possibility of data recovery in this scenario.
- **Lack of Auto-Lock:** The application remains unlocked as long as it is open. It does not automatically lock itself after a period of inactivity.
- **No Built-in Password Generator:** The current version requires users to create their own passwords rather than generating strong, random passwords natively.
- **Single User Profile:** The application supports only one vault per installation directory.

7 Conclusion

Password Vault Local (PVL) is a robust, lightweight, and secure solution for individuals who prefer offline data management. By leveraging industry-standard cryptographic primitives and modern GUI libraries, PVL ensures that user credentials are protected against unauthorized access while remaining easily accessible to the owner.